SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

Candidate Name

Class

CHEMISTRY **9647/01**

JC2 Preliminary Examination 28 Aug 2015

Paper 1 Multiple Choice 1 hour

Additional Materials: Data Booklet
Optical Mark Sheet (OMS)

READ THESE INSTRUCTIONS FIRST

On the separate multiple choice OMS given, write your name, subject title and class in the spaces provided.
Shade correctly your FIN/NRIC number.

There are **40** questions in this paper. Answer **all** questions.
For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the one you consider correct and record your choice using a **soft pencil** on the separate OMS.
Each correct answer will score one mark.
A mark will not be deducted for a wrong answer.

You are advised to fill in the OMS as you go along; **no additional time** will be given for the transfer of answers once the examination has ended.

Any rough working should be done in this question paper.

This document consists of **19** printed pages.

Answer all questions

1 Consider the following half-equations.

$Cr_2O_7^{2-}$ + $14H^+$ + 6e → $2Cr^{3+}$ + $7H_2O$
Fe^{2+} → Fe^{3+} + e
$2I^-$ → I_2 + 2e

Which of the following correctly reflects the volumes of 0.100 mol dm^{-3} of $K_2Cr_2O_7$ and 0.15 mol dm^{-3} of FeI_2 required for complete reaction?

	Volume of $K_2Cr_2O_7$ / cm^3	Volume of FeI_2 / cm^3
A	12.50	25
B	12.50	50
C	18.75	25
D	31.25	50

2 Long term exposure to radioactive isotope $^{226}_{88}Ra$ increases the risk of developing diseases such as bone cancer and leukaemia.

$^{226}_{88}Ra$ decays to give an element **A** and emits a high energy α-particle (which is a helium nucleus, $^{4}_{2}He$). No other particle is produced. α-Particles cause irreparable damage to the tissues of internal organs.

$$^{226}_{88}Ra \rightarrow {}^{4}_{2}He + \mathbf{A}$$

Which row in the table correctly describes the nuclear make-up of $^{226}_{88}Ra$ and **A**?

	$^{226}_{88}Ra$	**A**	
	number of neutrons	number of protons	number of neutrons
A	138	84	136
B	138	86	136
C	226	84	138
D	226	86	138

3 Beams of charged particles are deflected by an electric field. In an experiment, protons are deflected through an angle of $+15^{o}$.

A beam of particles, **F**, are separately passed through the same electric field and was deflected by an angle of $+5^{o}$. Each **F** particle has 12 times the mass of a proton.

What is the overall charge on a particle of **F**?

A 1+

B 2+

C 3+

D 4+

4 Which pair of compounds satisfies the following conditions?

(i) The first compound has a larger bond angle than the second compound.
(ii) The first compound is more polar than the second compound.

	First compound	Second compound
A	ClO_2	HCN
B	NF_3	SeF_6
C	IF_3	PH_3
D	$BeCl_2$	N_2H_4

5 The theoretical magnitude of lattice energy can be estimated from the Born-Landé equation, the relationship of which can be simplified as shown.

$$|\text{ lattice energy }| \propto \left|\frac{q_+ \times q_-}{r_+ + r_-}\right|$$

The consequence of the Born-Landé equation is that aluminium oxide has a higher theoretical magnitude of lattice energy as compared to magnesium oxide. However, the melting point of aluminium oxide is 2070 ^{o}C as compared to 2850 ^{o}C for magnesium oxide.

Which of the following could **not** be a possible reason for this anomaly?

A Aluminium oxide has a different lattice structure (i.e. different pattern of ions arrangement) as compared to magnesium oxide.

B Aluminium oxide has a more significant covalent character as compared to magnesium oxide.

C Aluminium ion has a much higher charge density as compared to magnesium ion.

D Aluminium has a stronger metallic bond as compared to magnesium.

6 Which of the following statements best explains why iodine is a solid at room temperature but water is a liquid?

A Iodine has a much larger molar mass than water.

B Iodine has strong and extensive covalent bonds between the atoms in a giant 3D structure.

C The van der Waals' forces of attraction between iodine molecules are stronger than the intermolecular hydrogen bonding of water.

D Water has intermolecular hydrogen bonding which are stronger than the van der Waals' forces of attraction between iodine molecules.

7 For a given mass of an ideal gas, which graph displays a different shape from the rest?

A PV against $\frac{1}{V}$ (at constant T)

B $\frac{V}{T}$ against T (at constant P)

C $\frac{density}{pressure}$ against P (at constant T)

D $\frac{1}{density}$ against T (at constant P)

8 For the reaction 2**B**(aq) + **C**(aq) $\longrightarrow$ **D**(aq) at room temperature, the rate equation is

$$\text{Rate} = k[H^+][\mathbf{B}]^2$$

Which statement is **not** correct?

A H^+ is a catalyst in the reaction.

B When the concentration of **C** is halved at 15°C, the rate decreases.

C When the concentrations of H^+ and **B** are doubled, the rate increases six times.

D The slow step of the reaction could be 2**B** (aq) + H^+ (aq) $\rightarrow$ **E** (aq), where **E** is an intermediate.

9 The pH of water varies with temperature according to the table below.

Temperature / ^{o}C	pH
25	7.0
62	6.5

Which of the following statements is true?

A The K_a is higher than K_b at 62 ^{o}C.

B Water becomes more acidic as temperature increases.

C The ionic dissociation of water is an exothermic process.

D The ionic product of water, K_w, increases at higher temperature.

10 The use of the *Data Booklet* is required for this question.

A voltaic cell is made up of Mg^{2+}/Mg half-cell and Fe^{3+}/Fe^{2+} half-cell.

Which one of the following statements is correct?

A The Mg^{2+}/ Mg half-cell is the positive electrode.

B Increasing the temperature has no effect on the e.m.f. of the cell.

C Addition of water to the Fe^{3+}/Fe^{2+} half-cell decreases the e.m.f. of the cell.

D Addition of aqueous sodium hydroxide to the Mg^{2+}/Mg half-cell decreases the e.m.f. of the cell.

11 20 cm^3 of 0.2 $moldm^{-3}$ of H_2SO_4 was mixed with 20 cm^3 of 0.2 $moldm^{-3}$ of NaOH. The reaction mixture increased by $\boldsymbol{x}$ ^{o}C.

Which of the following would give a similar temperature rise?

A 10 cm^3 of 0.4 $moldm^{-3}$ of H_2SO_4 was mixed with 20 cm^3 of 0.1 $moldm^{-3}$ of NaOH.

B 20 cm^3 of 0.2 $moldm^{-3}$ of H_2SO_4 was mixed with 20 cm^3 of 0.4 $moldm^{-3}$ of NaOH.

C 10 cm^3 of 0.1 $moldm^{-3}$ of H_2SO_4 was mixed with 10 cm^3 of 0.4 $moldm^{-3}$ of NaOH.

D 20 cm^3 of 0.1 $moldm^{-3}$ of H_2SO_4 was mixed with 20 cm^3 of 0.1 $moldm^{-3}$ of NaOH.

12 The use of the *Data Booklet* is required for this question.

Dinitrogen oxide burns in ethyne to produce water vapour, carbon dioxide and nitrogen gas as the only products.

$$5N_2O(g) + C_2H_2\ (g) \rightarrow H_2O(g) + 2CO_2(g) + 5N_2(g)$$

Given that the N=N and N=O bond energies in dinitrogen oxide is +418 kJ mol^{-1} and +686 kJ mol^{-1} respectively, what is the enthalpy change of the reaction?

A +1670 kJ mol^{-1}

B +1710 kJ mol^{-1}

C –1670 kJ mol^{-1}

D –1710 kJ mol^{-1}

13 Two elements **G** and **H** have the following properties.

- **G** and **H** form ionic compounds with the formula of Ca**G** and Ca**H**$_2$ respectively.
- **G** forms **GH**$_6$ molecule while **H** does not form **HG**$_6$.

Which pair of electronic configuration of **G** and **H** is correct?

(**G** and **H** do not represent elements in the periodic table)

	G	**H**
A	[He] $2s^2 2p^4$	[He] $2s^2 2p^2$
B	[He] $2s^2 2p^5$	[Ne] $3s^2 3p^2$
C	[Ne] $3s^2 3p^5$	[He] $2s^2 2p^5$
D	[Ne] $3s^2 3p^4$	[He] $2s^2 2p^5$

14 Which of the following statement is incorrect?

A Metal ions in aqueous solution behave as Lewis acids.

B Beryllium ion undergoes hydration and partial hydrolysis in water.

C As the size of the cation decreases, the magnitude of enthalpy change of hydration increases.

D The equilibrium constant for the hydrolysis of a hydrated cation is analogous to the K_a for the ionisation of a weak monoprotic acid.

15 **J**, **K** and **L** are 3 consecutive elements in period 3 of the periodic table.

The following information are given on the elements:

- **J** forms a cation.
- **K** has a higher melting point than **J**.
- **L** forms an anion.

Which of the following statement accounts for the above information?

A Electrical conductivity increases from **J** to **L**.

B The cationic radius of **J** is smaller than that of the cationic radius of **K**.

C The number of electrons in the cation of **J** is more than the cation of **K**.

D Effervescence is seen when a magnesium strip is dipped into the aqueous solution containing the chloride of element **L**.

16 Which solution will form a precipitate when aqueous silver nitrate is added?

A $Na_3[CrCl_6]$

B $[Cr(NH_3)_3Cl_3]$

C $[Cr(NH_3)_6]Cl_3$

D $[Cr(en)(NH_3)_2Cl_2]NO_3$

17 Which statement is **not** always true for transition metals and its ions?

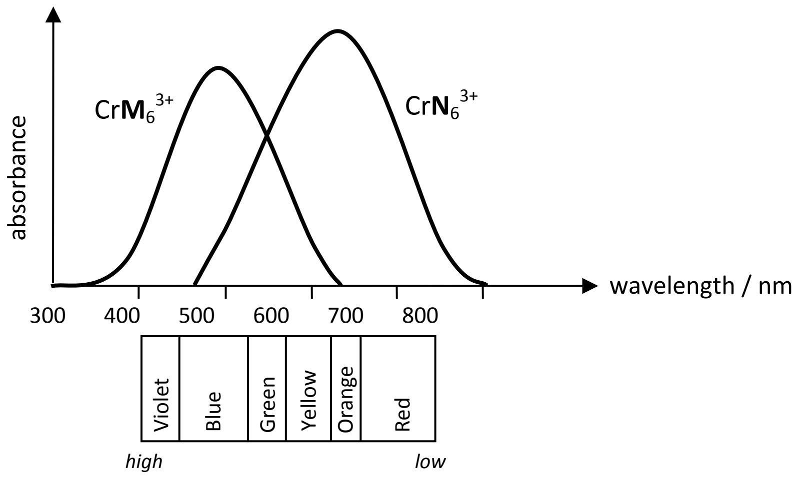
A The density of transition metals is greater than alkaline metals.

B Transition elements or their compounds are widely used as catalyst.

C The 3rd and 4th ionisation energy of transition metals increases sharply.

D The ionic radius of transition metals decreases across the period.

18 The diagram below shows the visible spectra of two chromium(III) complexes, $Cr\mathbf{M}_6^{3+}$ and $Cr\mathbf{N}_6^{3+}$. The various colours corresponding to the approximate wavelengths in the visible light region are shown below the axis.



What is the colour of each complex and which ligand causes a larger d-orbital splitting?

	colour of $Cr\mathbf{N}_6^{3+}$	colour of $Cr\mathbf{M}_6^{3+}$	Strength of field ligand
A	Red	violet	**M > N**
B	Red	violet	**N > M**
C	Blue	yellow orange	**M > N**
D	Blue	yellow orange	**N > M**

19 Covalent bonds are formed by orbital overlap. The shape of unsaturated hydrocarbon molecules can be explained in terms of hybridisation of orbitals.

Which bond is **not** present in CH_3CH_2CN?

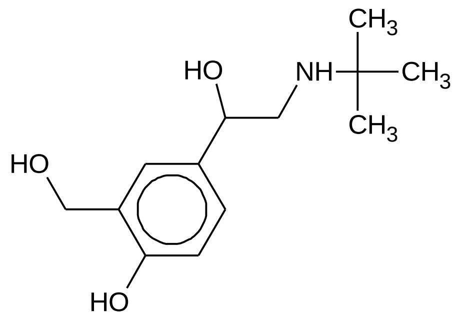
A a σ bond formed by 1s – $2sp^3$ overlap

B a σ bond formed by 2sp – $2sp^2$ overlap

C a σ bond formed by $2sp^3$ – $2sp^3$ overlap

D a π bond formed by 2p – 2p overlap

20 Salbutamol is a compound used in inhalers to relieve asthma.



It can be reacted with H_3PO_4 at $250^{o}C$.
What is the change in the number of stereoisomers after the reaction?

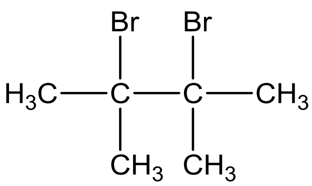
	Change in number of chiral carbon	Change in number of cis-trans isomers
A	Decrease of 1	Increase of 2
B	No change	Increase of 1
C	Decrease of 1	Increase of 1
D	No change	Increase of 2

21 How many structural isomers of carboxylic acids and esters are there in the formula $C_4H_8O_2$?

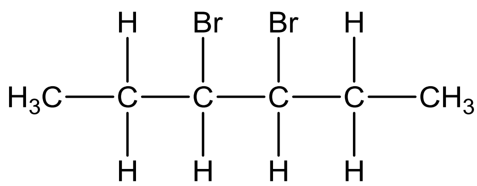
A 3 **B** 4 **C** 5 **D** 6

22 Which of the isomers of $C_6H_{12}Br_2$, on reaction with hot ethanolic potassium hydroxide for several hours, gives the greatest number of isomers?

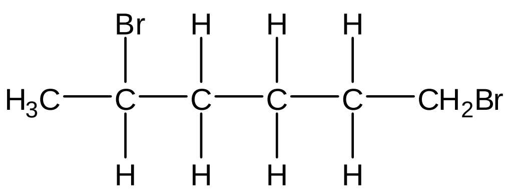
A



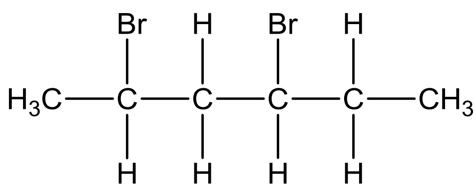
B



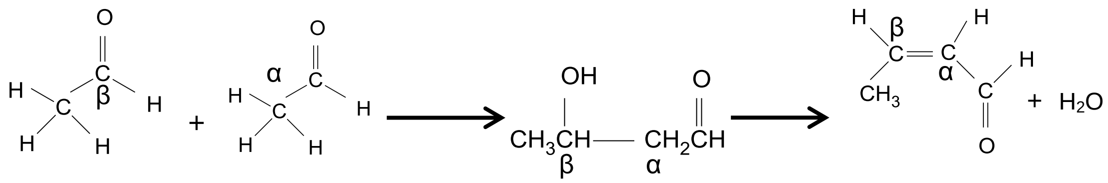
C



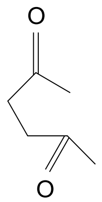
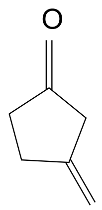
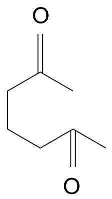
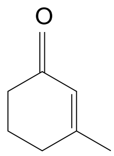
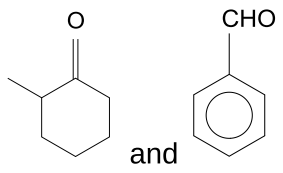
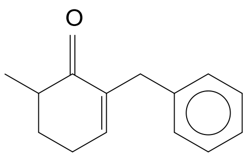
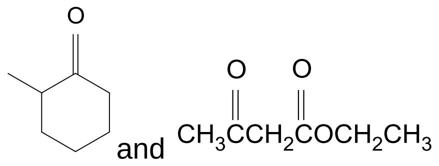
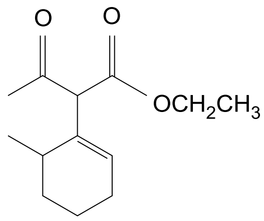
D



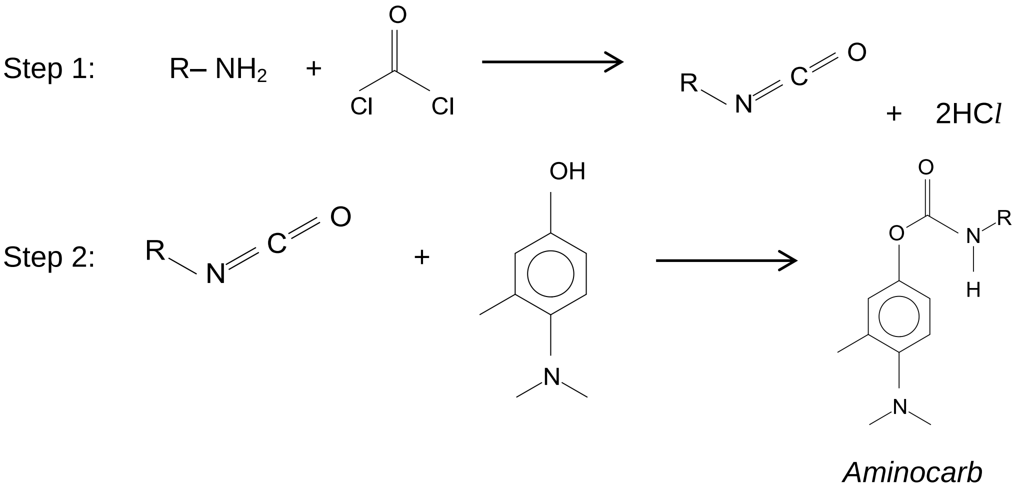
23 An aldol condensation is a reaction in which an enol or an enolate ion reacts with a carbonyl compound, followed by dehydration to give a conjugated enone. An example of the reaction can be seen from acetaldehyde below.



Which one of the following shows the correct product obtained from an aldol condensation?

	Reactants	**Product**
A		
B		
C		
D		

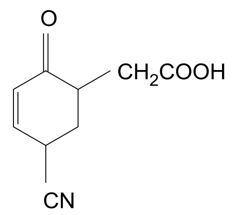
24 Aminocarb is an organic compound that is commonly used as a carbamate insecticide to protect cotton fields and forests from insect infestation. Aminocarb can be synthesized in two steps.



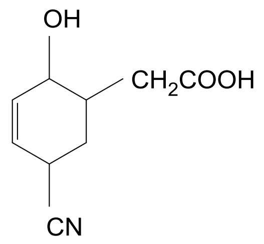
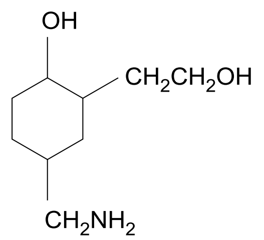
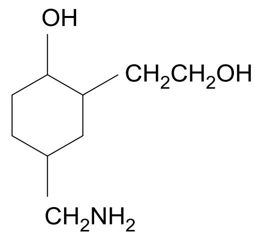
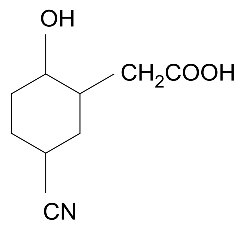
What are the types of reactions in steps 1 and 2?

	Step 1	Step 2
A	condensation	nucleophilic substitution
B	nucleophilic addition	electrophilic addition
C	condensation	nucleophilic addition
D	nucleophilic addition	electrophilic substitution

25 Compound **Q** can undergo reduction under suitable conditions.
Which of the following shows the correct reducing reagent and the product obtained?



Compound **Q**

	Reducing agent	Product
A	$NaBH_4$	
B	$LiAlH_4$, dry ether	
C	H_2, Pt catalyst	
D	H_2, Ni catalyst	

26 Which one of the following gives the compounds in order of decreasing acid strength?

A $CH_3CH_2COOH > CH_2C\mathit{l}CH_2COOH > CH_3CHC\mathit{l}COOH$

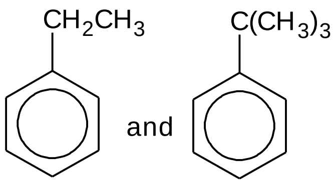
B $CH_2BrCOOH > CH_2C\mathit{l}COOH > CH_2FCOOH$

C $CF_3COOH > CHF_2COOH > CH_2FCOOH$

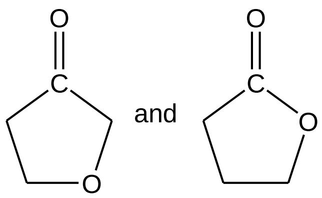
D $CH_3CH_2OH > C_6H_5OH > CH_3COOH$

27 Which pair of organic compounds **cannot** be distinguished by a chemical test?

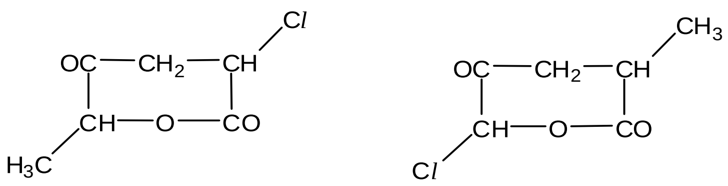
A



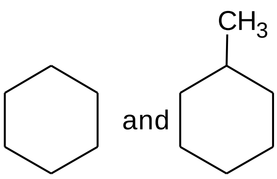
B



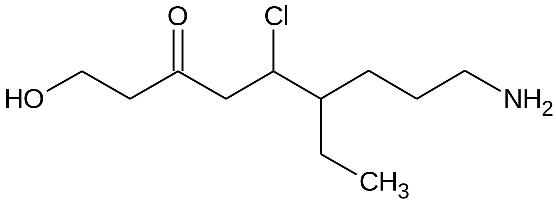
C



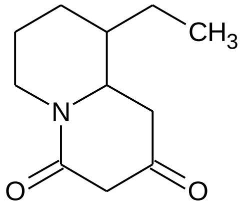
D

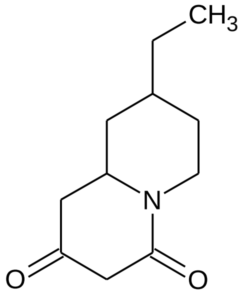


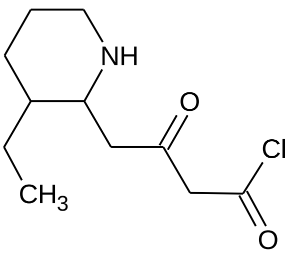
28

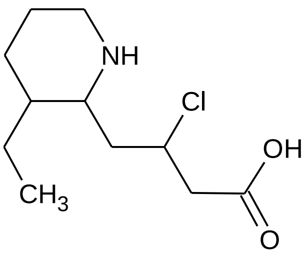


The compound above is reacted with acidified $K_2Cr_2O_7$, followed by PCl_5. The resulting compound undergone intramolecular reactions to form ringed structures under suitable conditions. Which structure is formed?

A 

B 

C 

D 

29 Partial hydrolysis of a polypeptide containing 7 amino acid residues using an enzyme gives the following fragments:

lys-glu-phe
ser-glu
phe-cys
glu-lys-glu
val-ser

What is the structure of the polypeptide?

A glu-lys-glu-phe-cys-val-ser

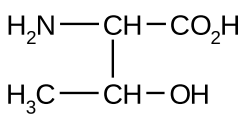
B val-ser-glu-lys-glu-phe-cys

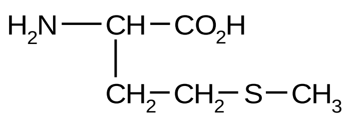
C ser-glu-lys-glu-phe-phe-cys

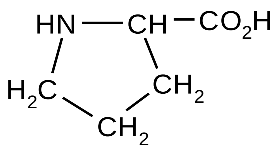
D glu-lys-glu-ser-glu-phe-cys

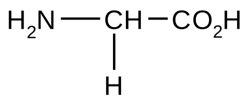
30 In the tertiary structure of a water-soluble globular protein extracted from barley grains, it was found that the types of amino acid residues present on the outer surface of the protein were different from those present on the inside.

Which of the following amino acids is most likely to be incorporated into the outer surface of such proteins?

A 

B 

C 

D 

For **questions 31 – 40**, one or more of the numbered statements **1** to **3** may be correct. Decide whether each of the statements is or is not correct. The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is to be used as correct response.

31 Which of the following statements about the nitrate(V) ion, NO_3^-, is correct?

1 The hybridisation of the nitrogen atom is sp^2.

2 NO_3^- has a total of twenty-four valence electrons.

3 All the nitrogen–oxygen bond lengths in NO_3^- are equal in length.

32 When an equilibrium is established in a reversible reaction, the standard Gibbs free energy, $\Delta G^{\ominus}_{eq}$, is related to the equilibrium constant, K_{eq}, by the following equation.

$$\Delta G^{\ominus}_{eq} = -2.303 \text{ RT} \log K_{eq}$$

where R is the gas constant and T is the temperature in Kelvin.

Which of the following statements are correct?

1 At constant temperature, a shift in position of equilibrium to the right results in the same value of $\Delta G^{\ominus}_{eq}$.

2 The reaction is spontaneous for all values of K_{eq}.

3 Adding a catalyst makes $\Delta G^{\ominus}_{eq}$ more negative.

33 The mechanism for a given reaction is shown below.

$$2\mathbf{S} \rightleftharpoons \mathbf{U} \qquad \text{fast}$$

$$\mathbf{U} + \mathbf{T} \rightarrow \mathbf{V} \qquad \text{slow}$$

$$\mathbf{V} + \mathbf{T} \rightarrow \mathbf{S}_2\mathbf{T}_2 \qquad \text{fast}$$

What conclusions can be drawn from the information given above?

1 Rate = $k[\mathbf{U}][\mathbf{T}]$

2 The overall equation is $2\mathbf{S} + 2\mathbf{T} \rightarrow \mathbf{S}_2\mathbf{T}_2$.

3 The rate of formation of $\mathbf{S}_2\mathbf{T}_2$ is proportional to the initial concentration of **T**.

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

34 The numerical values of the solubility product of calcium carbonate and calcium fluoride are 8.0×10^{-7} and 4.5×10^{-11} respectively at 30ºC.

Which of the following statements are correct?

1 Calcium carbonate is more soluble than calcium fluoride.

2 Addition of calcium nitrate solution to a solution containing equal concentration of carbonate and fluoride ions causes calcium carbonate to precipitate out first.

3 The solubility product of calcium carbonate decreases in a solution containing calcium nitrate.

35 When drops of sodium hydroxide were added to a green solution of chromium(III) nitrate, a grey-green precipitate was formed. The precipitate dissolved when excess sodium hydroxide was added, forming a dark green solution. Subsequent additions of liquid ammonia caused the solution to turn violet.

According to the information given above, which of the following statements is correct?

1 Hydroxide ions act as ligands in the first addition of sodium hydroxide.

2 When liquid ammonia is added, ligand exchange occurs.

3 The equilibrium constant, K_{stab} for the formation of the ammonia-containing complex is larger than that of the hydroxide-containing complex.

36 A catalytic converter is part of the exhaust system of modern cars.

Which reactions occur in a catalytic converter?

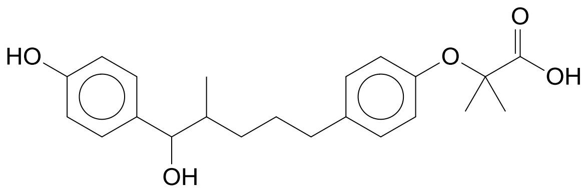
1 $C_xH_y + (x + y/4)O_2 \rightarrow xCO_2 + (y/2)\ H_2O$

2 $2CO + 2NO \rightarrow 2CO_2 + N_2$

3 $2SO_2 + 2NO \rightarrow 2SO_3 + N_2$

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

37



Z

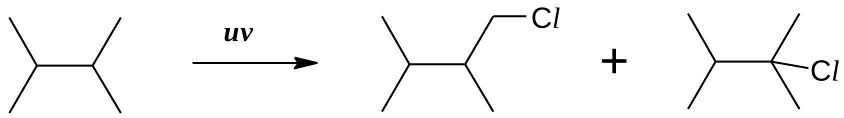
Which of the following statements about the reactions of **Z** is **incorrect**?

1 1 mol of **Z** reacts with $SOCl_2$ to release 3 mol of HCl gas.

2 1 mol of **Z** reacts with Na metal to release 3 mol of H_2 gas.

3 1 mol of **Z** reacts with Na_2CO_3 to release 2 mol of CO_2 gas.

38 2,3–dimethylbutane reacts with chlorine under *uv* to yield two monochlorinated products as shown in the reaction below.



product ratio is as follows 55 : 45

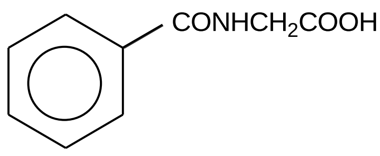
Which of the following statements explain the resultant product ratio?

1 There are more primary hydrogen atoms than tertiary hydrogen atoms.

2 The intermediate tertiary radical is more stable.

3 H• is not formed in the reaction mechanism.

39 Benzoylglycine (hippuric acid) was first isolated from a urine sample.



Which properties does this compound possess?

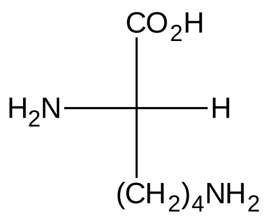
1 It can be hydrolysed to produce an amino acid.

2 It can be made by reacting benzoyl chloride with aminoethanoic acid.

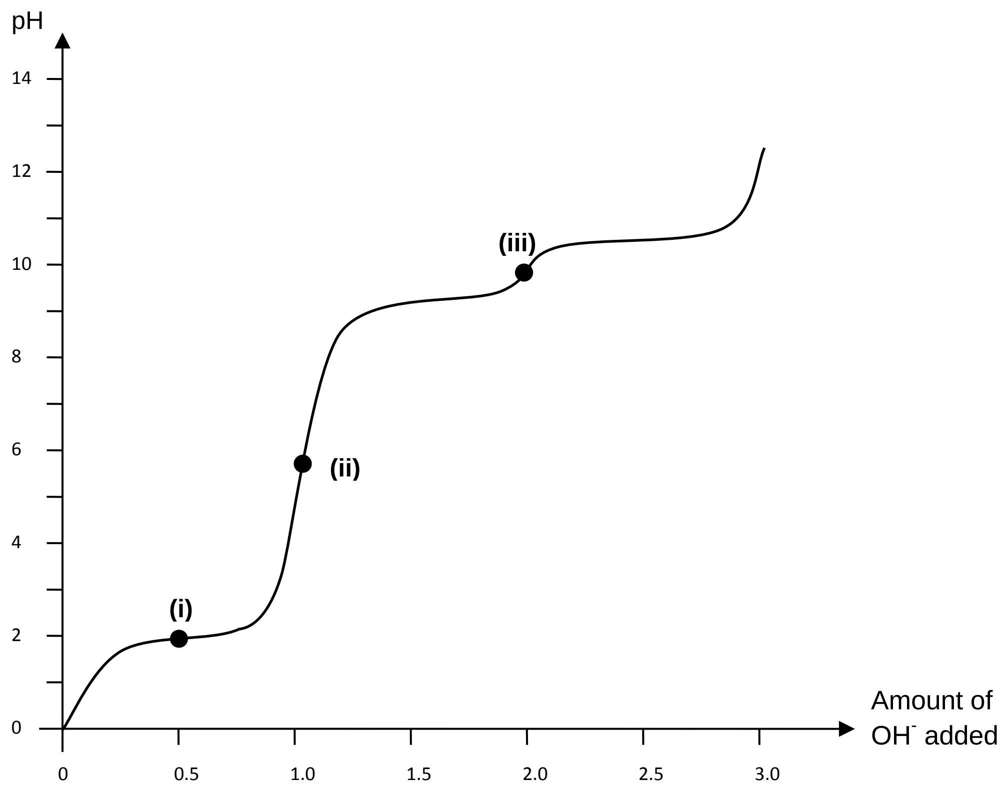
3 It can be neutralised by reaction with cold aqueous sodium hydroxide.

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 Lysine is an essential amino acid found in the body. It has three pK_a values associated with it: 2.2, 9.0 and 10.5.



When one mole of protonated lysine was titrated against hydroxide ions, the following pH curve is obtained:



Which of the following statements are true with respect to the curve above?

1 The major species present at point **(iii)** has no net charge.

2 The major species present at point **(ii)** will migrate towards the cathode of an electrolytic cell.

3 Equal amounts of $H_3N^+CH(CO_2H)(CH_2)_4NH_3^+$ and $H_3N^+CH(CO_2H)(CH_2)_4NH_2$ are present at point **(i)**.

END OF PAPER 1